ShiftOps AI

Intelligent Workforce Recovery • A human-in-the-loop response to staffing disruptions

From sudden absence to a safe, documented schedule change

An operational decision-support prototype that combines availability and roster checks, candidate eligibility, explainable recommendations, manager approvals, and audit records.

THE RESPONSE LOOP

- **Detect staffing gap**
- **Validate candidates**
- **Review a proposal**
- **Apply + audit change**

LIVE PROTOTYPE shiftops-ai-web.onrender.com • Synqvera Labs • October 2026

A single absence creates a chain of operational decisions

A shift gap is not simply an empty calendar slot; managers must balance coverage, rules, and accountability.

Coverage delays

Finding an available, role-matched replacement often involves checking separate records and contacting several employees.

Assignment risk

An apparently free employee may be on leave, have an overlapping shift, or exceed configured work limits.

Weak traceability

Without a structured decision trail, it is difficult to show why someone was selected or who approved the change.

Operational consequence

Slow response → inconsistent decisions → avoidable scheduling errors → hard-to-audit outcomes

DESIGN GOAL Support faster decisions without sacrificing staffing rules or manager control.

User needs drive product requirements

The concept is organised around the work managers and employees need to complete—not around a chatbot interface.

Stakeholder	Operational need	ShiftOps AI response
Operations manager	Resolve coverage gaps with confidence	Rank eligible options and review proposals
Roster coordinator	Maintain an accurate operational schedule	See shifts, availability, leave and status
Employee	Know assignments and record availability	Employee-facing schedule and availability views
Compliance / owner	Explain and reconstruct decisions	Manager approval and auditable changes

KEY REQUIREMENT Recommendations must be understandable, authorised, and based on actual scheduling data.

One controlled incident-to-resolution workflow

Each stage has a clear responsibility and a visible decision boundary.

01 **Identify gap** › 02 **Evaluate** › 03 **Explain** › 04 **Propose** › 05 **Approve** › 06 **Apply + audit**

The language model explains pre-validated information; it does not grant eligibility, approve a proposal, or write the roster.

Automated assessment

Status, role, availability, leave, conflicts, and weekly limits are checked by backend rules.

Human-governed action

A manager explicitly approves or rejects. Eligibility is rechecked before a transactional update.

DISTINCTION Useful AI assistance without autonomous schedule mutation.

The core product: connected workforce operations

A cohesive web application supports daily planning and urgent staffing recovery.

Employee management

Profiles, role, status, detail pages and reactivation

Availability

Recurring weekly windows and unavailable periods

Shift management

Create, edit and cancel a shift with state clarity

Weekly roster

Day, week, month and custom date filtering

Leave management

Requests, manager review and approved absences

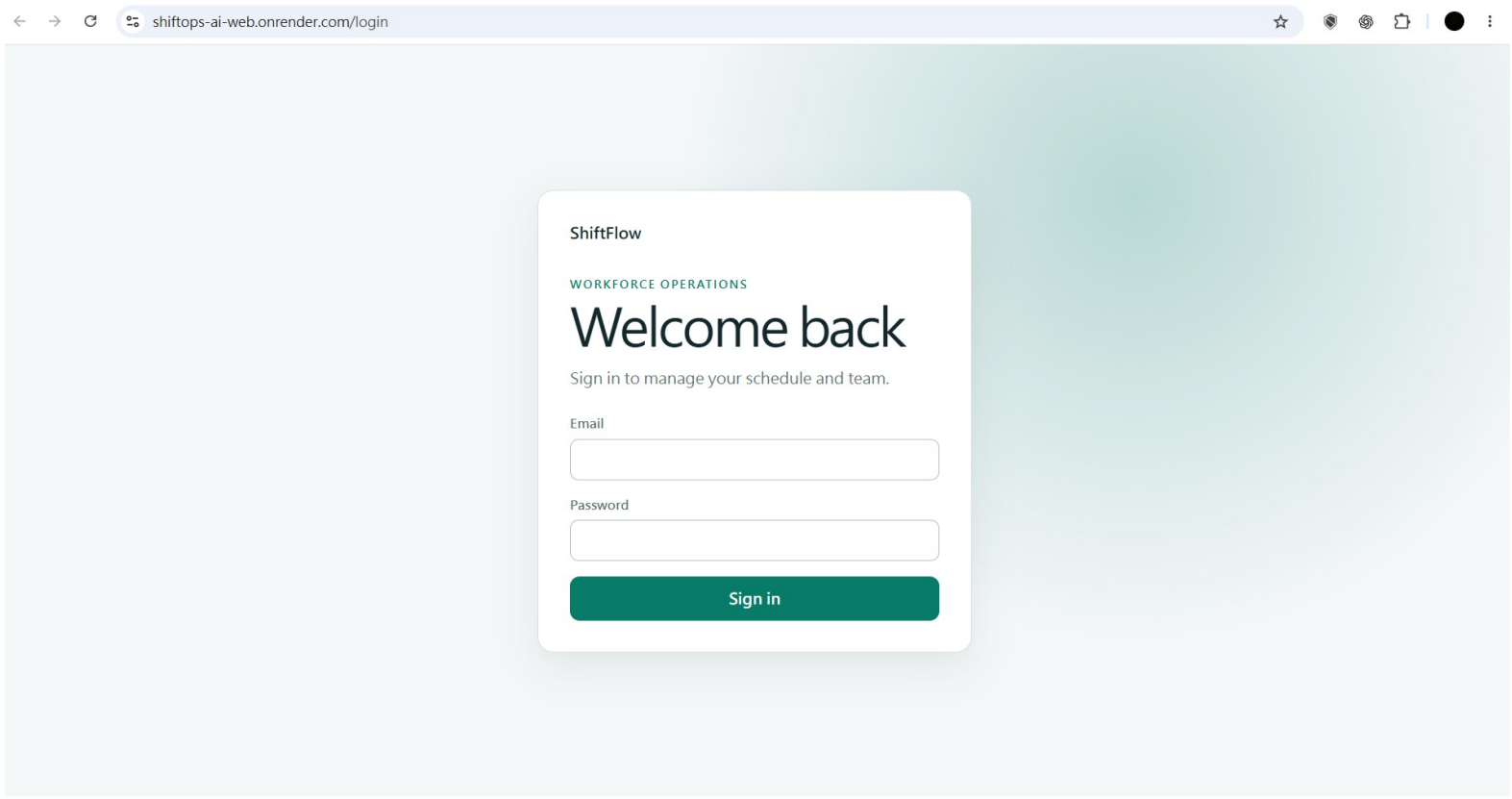
ShiftOps Copilot

Incident, eligibility, proposal, approval and audit

BUILT FOR USE A manager can move from staff records to a validated roster change within one application.

Public demo: real deployed ShiftOps AI interface

Screenshot evidence from the live React application; subsequent diagrams illustrate real system workflows.



Access and roles

Manager and employee authentication uses JWT and role-based access control.

Public demo

Frontend: shiftops-ai-web.onrender.com
API: shiftops-ai-api.onrender.com
Database: PostgreSQL

Employee and availability management

Reliable replacement suggestions depend on accurate source records and clear access controls.

Module	Manager interaction	Decision value
Employee profiles	Review name, job title, status and contact data	Clarifies assignment and qualification context
Status and reactivation	Deactivate / reactivate an existing employee	Avoids assigning inactive workers
Weekly availability	View Monday–Sunday working windows	Excludes periods a person cannot cover
Leave requests	Review approved or pending requests	Prevents conflicts with approved leave
RBAC	Separate Manager and Employee permissions	Restricts staffing changes to authorised users

WHY IT MATTERS Clean workforce data lets the eligibility engine reject unsuitable candidates before AI explanation.

Shift management and flexible roster views

The roster provides the operational ground truth for each staffing decision.

Create and maintain shifts

Managers create shifts with dates, times and required roles, then edit or cancel existing assignments. Clear form states reduce accidental changes.

Search time periods

The UI supports day, week, month and custom start/end dates, with period navigation and empty states based on real API data.

ILLUSTRATIVE ROSTER DATA FIELDS

- **Date + weekday**
- **Time window**
- **Required role**
- **Assigned employee**
- **Shift status**

UX PRINCIPLE Display real schedule data; do not invent published or historical shifts.

Eligibility rules protect the recommendation

The backend makes the pass/fail decision; explanation is downstream.

Validation	What the system asks	Why it protects operations
Employment status	Is the employee active?	Inactive employees cannot be considered
Role match	Can this employee perform the required role?	Avoids capability mismatch
Availability	Is the employee available for this time?	Prevents unavailable assignments
Approved leave	Does requested coverage overlap leave?	Respects approved absence
Shift overlap	Is another shift already assigned?	Prevents competing roster commitments
Weekly limits	Would scheduled hours exceed configured limits?	Enforces workload constraints

SAFETY GUARANTEE A fluent AI explanation never overrides a rejected eligibility check.

Where AMD Token Factory fits in the architecture

Grounded information is separated from model-generated language and scheduling authority.

01 **Validated facts** › 02 **AMD adapter** › 03 **Explanation** › 04 **Manager review**

The AMD model interface is used to create natural-language explanations when remote inference is available. The application can fall back to deterministic explanation.

Verified engineering capability

AMD-compatible request adapter and strict verification path are implemented. The documented model identifier was MiniCPM5-2B.

Evidence boundary

Public deployment was initially configured as deterministic. Live AMD inference must be evidenced separately; a successful fallback is not remote inference.

RESPONSIBLE CLAIM No unverified GPU execution, latency improvement, or model accuracy figures are claimed.

Manager approval, safe application and audit

A recommendation only becomes a roster change after explicit authorisation and final checks.

1	**Proposal**	Records the suggested replacement and rationale; no schedule mutation
2	**Manager decision**	Approves or rejects the proposal through a controlled API
3	**Revalidation**	Checks eligibility again to guard against stale staff/shift state
4	**Atomic apply**	Applies eligible shift changes within a transaction
5	**Audit record**	Persists the decision and resulting operational event

CONTROL DESIGN Managers remain accountable; AI output cannot commit a scheduling change.

Architecture: real front end, APIs, database and AI adapter

Designed around widely used web technologies and clear separation of concerns.

WEB FRONTEND

React + TypeScript + Vite

Theme, views, forms, navigation

API + SECURITY

Express + Node.js + JWT

RBAC, input validation, workflows

DATA + RULES

PostgreSQL + Prisma

Schedule records, eligibility, transactions

AI EXPLANATION

AMD Token Factory adapter

Grounded remote inference / fallback

Deployment and quality

Render Static Site + Web Service + PostgreSQL; Prisma migrations and demo seed; Vitest, Supertest, TypeScript, ESLint and CI.

BOUNDARY Frontend requests operations; backend enforces rules and controls every state-changing action.

Business value: measurable questions worth testing

The prototype addresses operational outcomes but does not claim measured production ROI.

Business objective	How the design helps	Suggested pilot measure
Reduce time to coverage	Pre-filter candidates and centralise the incident flow	Minutes from gap detection to approved coverage
Lower assignment risk	Verify conflicts, roles, leave and weekly limits	Invalid-assignment attempts prevented
Improve consistency	Apply the same rules across each candidate	Share of recommendations with explicit rule reasons
Strengthen accountability	Capture approvals, changes and audit records	Percentage of changes with complete decision history
Support manager trust	Explain validated recommendations and allow rejection	Manager acceptance and override frequency

EVIDENCE STANDARD Potential benefits are hypotheses for future pilots—not demonstrated customer savings.

Why build ShiftOps AI when workforce platforms exist?

Traditional scheduling products validate the need; this prototype emphasises incident recovery governance.

Existing workforce tools

Scheduling, leave administration, shift swaps and notifications are common capabilities in established platforms. Their existence supports the practical relevance of this problem.

ShiftOps AI focus

A clear incident → eligibility → explanation → approval → apply → audit pathway with deterministic safeguards and explicit human control.

POSITIONING

Not a replacement for enterprise workforce management suites today: a focused, testable workflow for accountable disruption response that can complement broader platforms.

NEXT VALIDATION Interview real operations managers and measure incident resolution in a controlled pilot.

What has been delivered—and what comes next

Separate verifiable prototype achievements from future enhancements and unverified AI claims.

Workstream	Current evidence	Next step
Core workflow	Implemented through Phase 10; reported automated tests pass	Record public end-to-end walkthrough
Public demo	React, Express and PostgreSQL deployed to Render	Capture product screenshots / video
UI accessibility	Theme, clearer buttons and structured management views	Manual usability feedback
AI integration	AMD adapter implemented; latest live evidence to confirm	Verify inference without fallback
Business validation	Operational value proposition and working prototype	Pilot metrics and user interviews

SHORT-TERM PRIORITY Complete the hackathon video, pitch materials, live AMD verification and final submission.

ShiftOps AI — ready to explore

An accountable way to respond to staffing disruptions, from validated options to audited action.

TRY THE PUBLIC DEMO

shiftops-ai-web.onrender.com

VIEW THE CODE

github.com/123114-robot/shiftops-ai-act3

TEAM Synqvera Labs • AMD Developer Hackathon ACT III • October 2026